

Quiz2, MAT1375 Professor Chiu

ID: _____

Name: Sol.

- This quiz consists of 3 questions for a total of 10 points.
- You have 15 minutes to complete the quiz.
- Show all work and justify your answers.
- Wishing you success.

1. Let f be the piecewise defined function:

$$f(x) = \begin{cases} 2x + 3, & \text{for } -8 < x \leq 4 \\ \frac{x}{2}, & \text{for } -3 < x < 2 \\ x^2 + 5x, & \text{for } 4 \leq x \end{cases}$$

Find $f(6)$, $f(2)$, $f(-6)$, and $f(100)$.

$$f(6) = 6^2 + 5 \cdot 6 = 36 + 30 = 66$$

$\downarrow$
 $x \geq 4$

$$f(2) = 2 \cdot (2) + 3 = 7$$

$\downarrow$
 $-8 < x \leq 4$

$$f(-6) = 2 \cdot (-6) + 3 = -9$$

$\downarrow$
 $-8 < x \leq 4$

$$f(100) = 100^2 + 5 \cdot 100 = 10000 + 500 = 10500$$

$\downarrow$
 $x \geq 4$

2. Given a function $f(x) = \frac{x+1}{x^2-7x+10}$. Find the domain of f . *Hint: find the factor decomposition of denominator.*

$$f(x) = \frac{x+1}{x^2-7x+10} = \frac{x+1}{(x-2)(x-5)}$$

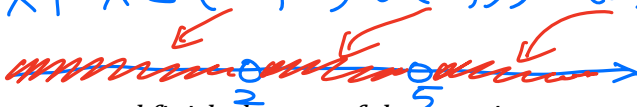
$\downarrow$

a fraction is "not good" when its denominator is zero.

$$x^2 - 7x + 10 = (x-2)(x-5) \Rightarrow (x-2)(x-5) \neq 0 \Rightarrow (x-2) \neq 0, (x-5) \neq 0$$

$\Rightarrow x \neq 2, x \neq 5$

$$\text{Domain of } f : D_f = \{x \in \mathbb{R} \mid x \neq 2, x \neq 5\} \text{ or } \{x \mid x \in (-\infty, 2) \cup (2, 5) \cup (5, \infty)\}$$



Please turn over and finish the rest of the question.

3. Given a function $f(x) = x^2 - 2x + 5$. Find (a) $f(-2)$; (b) $f(x + h)$; (c) $\frac{f(x+h)-f(x)}{h}$.

$$\begin{aligned} \text{(a)} \quad f(-2) &= (-2)^2 - 2(-2) + 5 \\ &= 4 + 4 + 5 = 13 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad f(x+h) &= (x+h)^2 - 2(x+h) + 5 \\ &= x^2 + 2xh + h^2 - 2x - 2h + 5 \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad f(x+h) - f(x) &= (x^2 + 2xh + h^2 - 2x - 2h + 5) - (x^2 - 2x + 5) \\ &= x^2 + 2xh + h^2 - 2x - 2h + 5 - x^2 + 2x - 5 \\ &= 2xh + h^2 - 2h \end{aligned}$$

$$\frac{f(x+h) - f(x)}{h} = \frac{2xh + h^2 - 2h}{h} = \frac{h(2x + h - 2)}{h} = 2x + h - 2$$

End of this quiz.